Course Code: CSE 3107

Course Title: Communication Engineering

Dr. Foez Ahmed, SMIEEE

Associate Professor | Dept. of ICE

University of Rajshahi, Rajshahi-6205

Email: foez28@ru.ac.bd

Lecture # W12

Satellite Communication

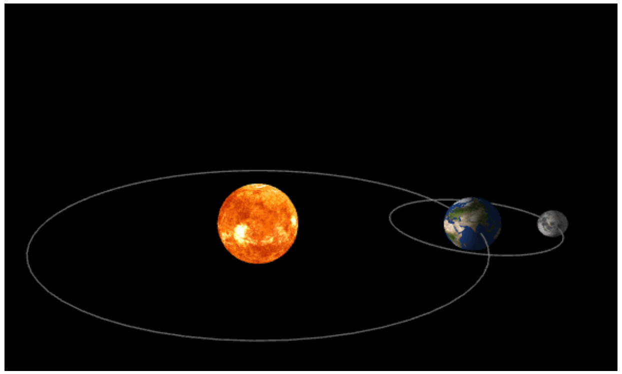
Outline

- Satellite & Satellite Communication,
- Kepler's First and Second Law,
- Orbits and Geostationary Orbits

Satellite Communication

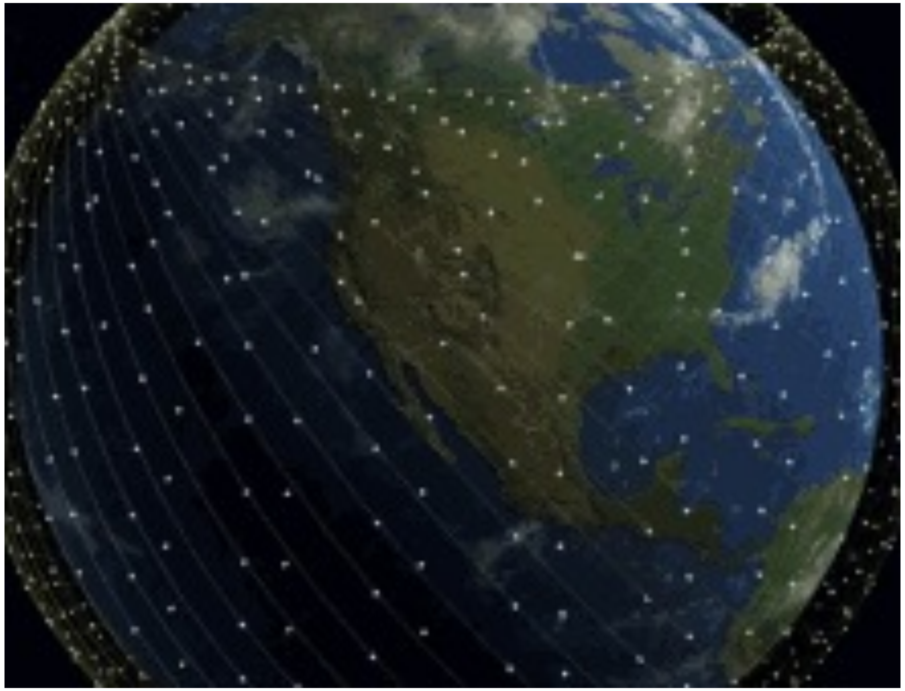
- **Satellite?**
  - ✓ A satellite is an object that moves around a larger object in space, Or
  - ✓ A satellite is any object that orbits a planet or star in a fixed, repeating path in space.

- There are generally two types of satellite.
  - ✓ Natural Satellite and Artificial Satellite (Man Made Satellite).



- **Natural Satellite**: A natural satellite is a celestial body, such as a moon, planetesimal, or asteroid, that orbits a planet, dwarf planet, or smaller body in space. These bodies are not man-made and are held in orbit by gravitational attraction. Examples: Earth and the moon

Satellite Communication



- **Artificial Satellites:**

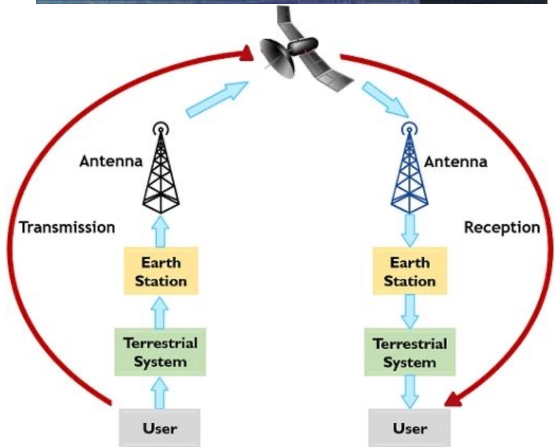
  These machines—ranging from small satellites to the massive International Space Station—are launched by rockets to perform functions like communications, weather forecasting, navigation (GPS), and scientific research etc.

- **Examples:**

  Intelsat satellites and Starlink satellites - Used for transmitting television, telephone, radio, and internet signals.

Satellite Communication

- **Satellite communication**
  - ✓ It is an advanced communication technology which involves the transmission of information from one point to another by using an artificial satellite orbiting the Earth.
  - ✓ The satellite is known as a communication satellite and it is a specially designed artificial satellite that can create a communication link to transmit signals between transmitter and receiver on the Earth.
  - ✓ Satellite communication is an effective communication method for transmitting information over very large distances across the globe.



Structure of Satellite Communication

Satellite Communication

- ✓ In satellite communication, the artificial satellite in the orbit of the earth acts as a relay station. It receives signals from one location on the earth and retransmits them to another location on the earth. Examples of satellite communication include GPS navigation and remote internet access.

- ✓ Satellite communication is serving as the backbone of internet and global telecommunication services.

- ❑ There are two main components of a typical satellite communication −
  1. Ground segment − It consists of devices and equipment on the earth such as transmission system, receiving devices, and ancillary equipment.
  2. Space segment − It mainly consists of communication satellites orbiting the earth.

- ❑ Satellite communication takes place through uplinking and downlinking of signals between the earth's station and the satellite.

❑ Block diagram of Satellite Communication:

- ✓ **Transmitter (Transmitting Earth Station)** − It is a device located at a point on the earth and it sends information signals to the satellite.
- ✓ **Satellite Transponder (Satellite)** − It is a device located on the communication satellite in the space. It receives information signals from the earth station, amplify and retransmits them to another location on the earth.

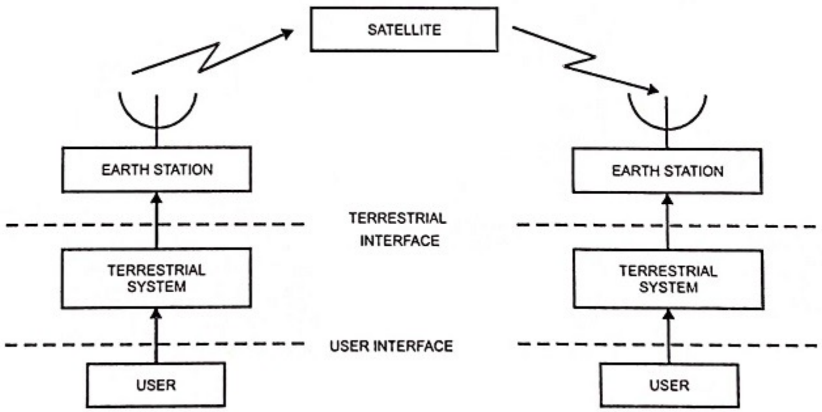


Fig. 24.7 *Basic Structure of a Satellite Communication System*

- ✓ **Receiver (Receiving Earth Station)** − It is the destination device in the satellite communication, located on the earth. It captures the retransmitted data from the satellite and process them.

❑ How Does Satellite Communication Work?

- ✓ Satellite communication completes its process of information transmission between two locations on the earth in the following three simple steps −

Step 1 − First of all, a device or station on the earth sends information signals to the communication satellite in the space. This process is called **uplinking**.

Step 2 − The satellite transponder processes the uplinked signals to amplify them.

Step 3 − The processed signals are retransmitted by the satellite transponder back to a device or station at another location on the earth. This process is called **downlinking**.

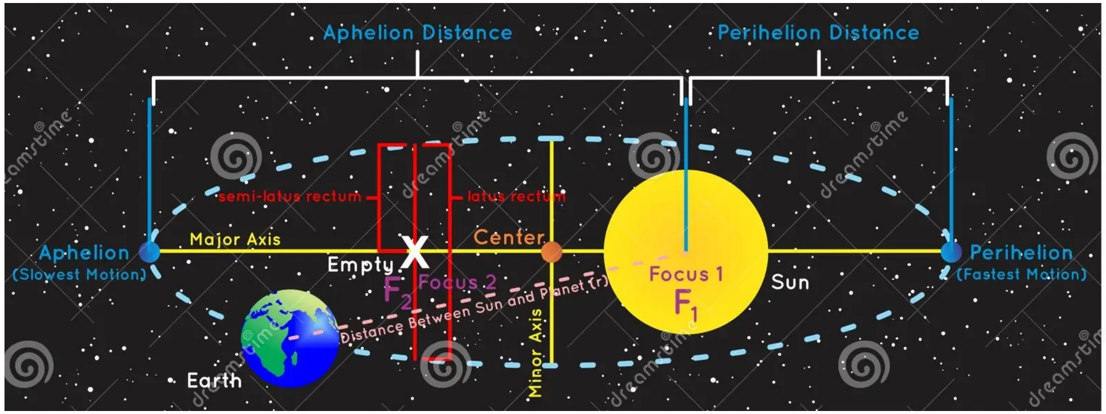
- **Frequency Bands used in Satellite Communication:**

Band	Frequency Range	Applications
L-band	1–2 GHz	Mobile satellite
S-band	2–4 GHz	Weather radar
C-band	4–8 GHz	TV broadcasting
X-band	8–12 GHz	Military
Ku-band	12–18 GHz	DTH television
Ka-band	26–40 GHz	High-speed internet

Kepler's Laws of Planetary Motion:

First Law of Planetary Motion:

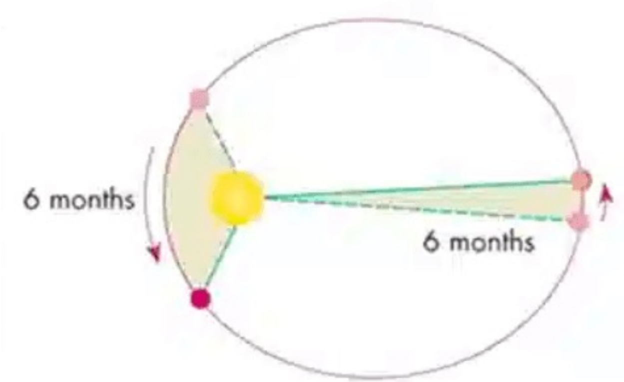
- Statement: "***Every planet moves around the Sun in an elliptical orbit, with the Sun located at one of the two foci of the ellipse***."
- The first law of planetary motion is also called the Law of Orbit.



- In simpler terms, planets do not move in perfect circles but rather follow elliptical paths with the Sun at one of the focal points.
- The point at which the planet is closest to the sun is called the **perihelion**, and the point at which the planet is farthest from the sun is called **aphelion.**

❑ Kepler's Second Law of Planetary Motion

- ✓ Statement: "**The radius vector that is drawn from the sun to the planet sweeps out equal areas in equal intervals of time**".
- ✓ The second law of Kepler's law of Planetary Motion is also called the "**Law of Equal Areas**".
- ✓ In other words, a planet moves faster when it is closer to the Sun and slower when it is farther away.
- ✓ This law highlights the fact that planets do not move at a constant speed throughout their orbits.

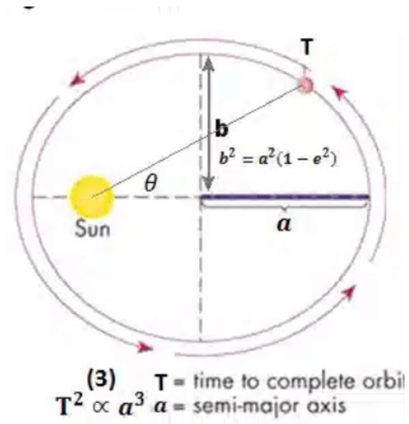


(2)
Equal areas in equal time

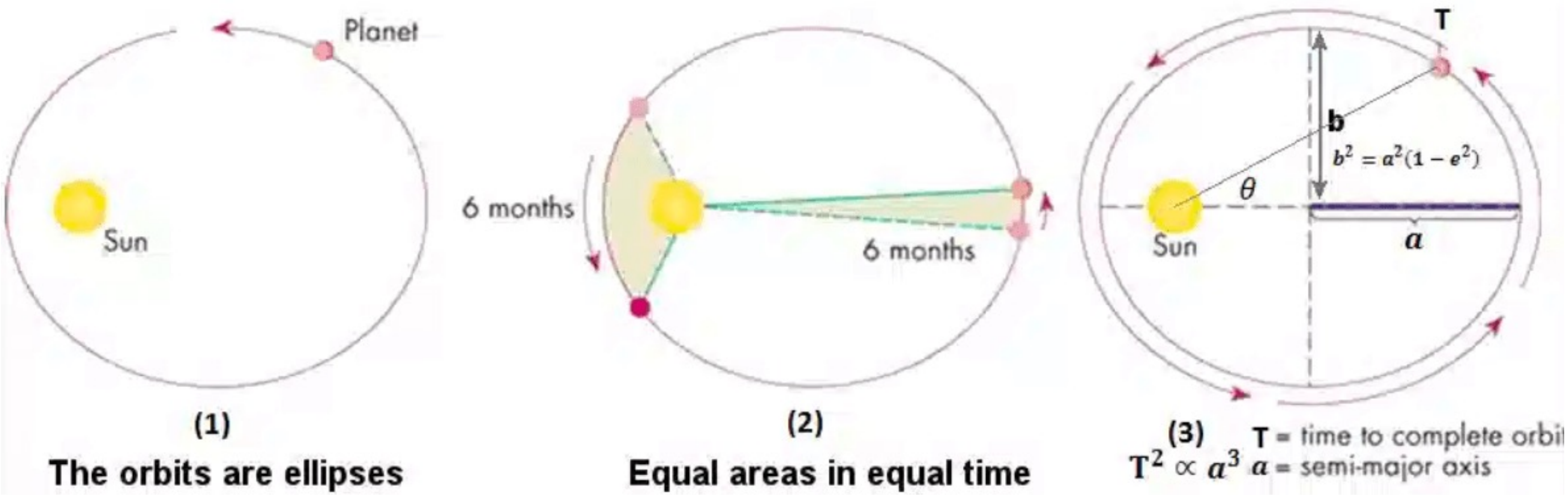
- **Third (3^{rd}) Law**:
  - The third law of Kepler's is also known as the Law of Harmonies and the Law of Periods.
  - Kepler's Third Law establishes a mathematical relationship between a planet's orbital period and its average distance from the Sun.
  - According to this law, "***The square of the time period of revolution of a planet (T) is proportional to the cube of the semi-major axis of the ellipse traced out by the planet***"

$$T^2 \propto a^3$$

  - Where, *T* is the time period of the planet and *a* is the semi-major axis .
  - Kepler's Third Law plays a crucial role in determining the <u>relative sizes and distances of planets</u> within our solar system.

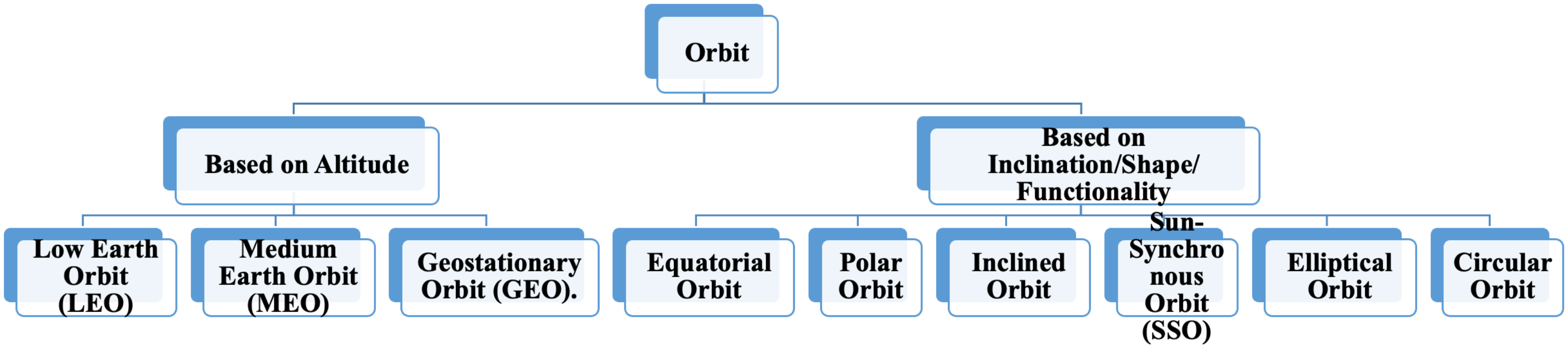


Kepler's 3 Laws of Planetary Motion



❑ Orbits:

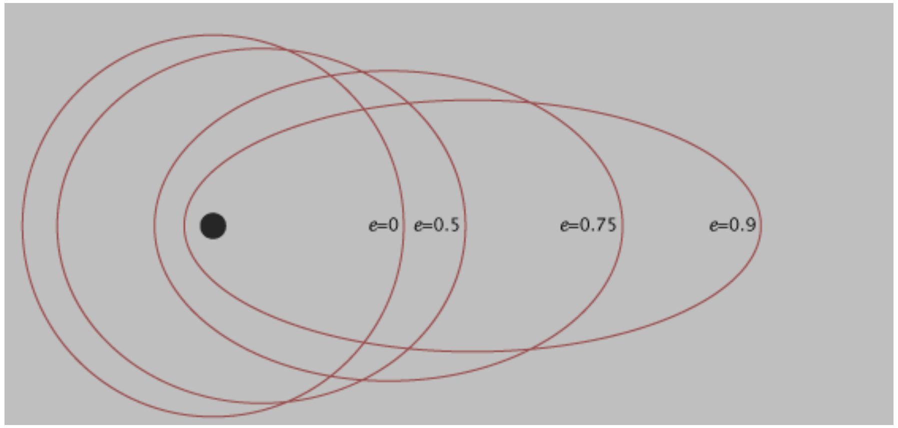
- ✓ An orbit is the regular path along which a planet, moon, or satellite moves around another object in space.
- ✓ The type of orbit is determined by the satellite's altitude, shape, inclination and functionality (its purpose, such as communication, observation, or scientific research)



❑ Eccentricity of Orbit

- ✓ Eccentricity refers to the shape of the orbit. A satellite with a low eccentricity orbit moves in a near circle around the Earth.

- ✓ The eccentricity (e) of an orbit indicates the deviation of the orbit from a perfect circle.



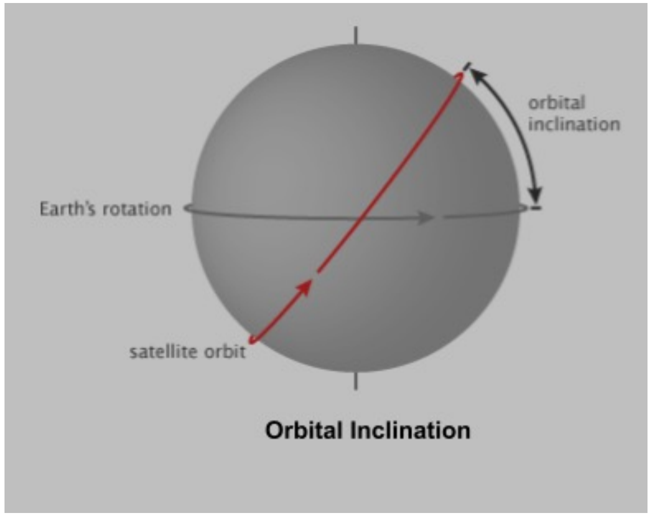
- ✓ A circular orbit has an eccentricity of 0, while a highly eccentric orbit is closer to (but always less than) 1.

- ✓ A satellite in an eccentric orbit moves around one of the ellipse’s focal points, not the centre.

❑ Inclination of Orbit

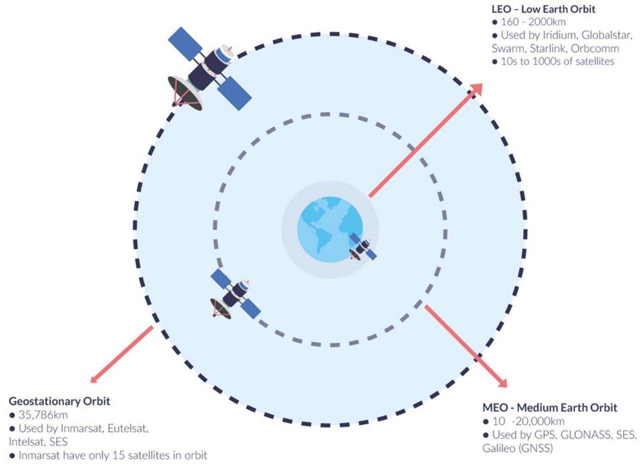
- ✓ The inclination is the angle of the orbit in relation to Earth's equator.

- ✓ An orbital inclination of 0° is directly above the equator, 90° crosses right above the pole, and 180° orbits above the equator in the opposite direction of Earth's spin.



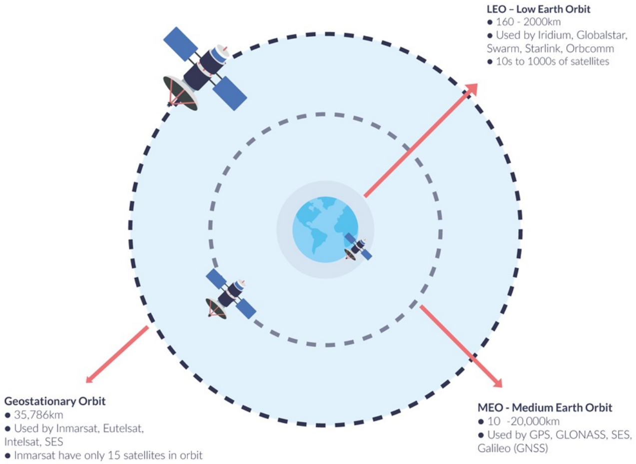
Satellite Communication

Orbit Type	Low Earth Orbit (LEO)
Altitude	160 km to 2,000 km
Orbit Characteristics	Satellites in LEO are closest to Earth, orbiting in about **90 minutes**. Used for weather monitoring, Earth observation, and communication.
Usage	The International Space Station (ISS), Earth observation satellites, communication satellites
Advantages	Low latency, high-resolution images of Earth
Disadvantages	Shorter lifespan due to atmospheric drag, must be repositioned often

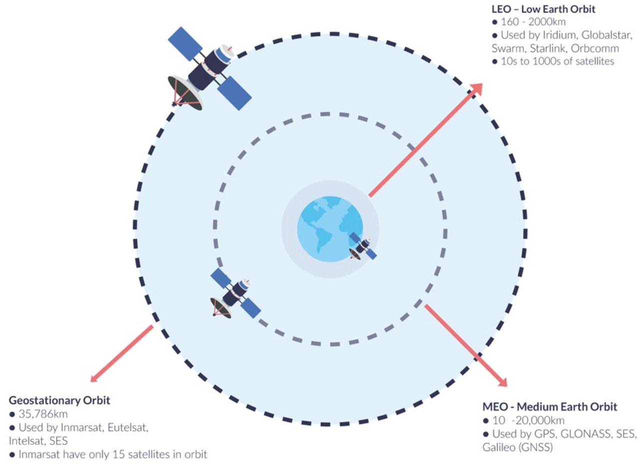


Satellite Communication

Orbit Type	**Medium Earth Orbit (MEO)**
Altitude	2,000 km to 35,786 km
Orbit Characteristics	MEO satellites orbit at higher altitudes (2-24 hours per orbit). Used for navigation systems.
Usage	Global navigation systems (GPS, Galileo, GLONASS)
Advantages	Longer lifespan compared to LEO, larger coverage area
Disadvantages	Higher communication delay compared to LEO due to increased distance

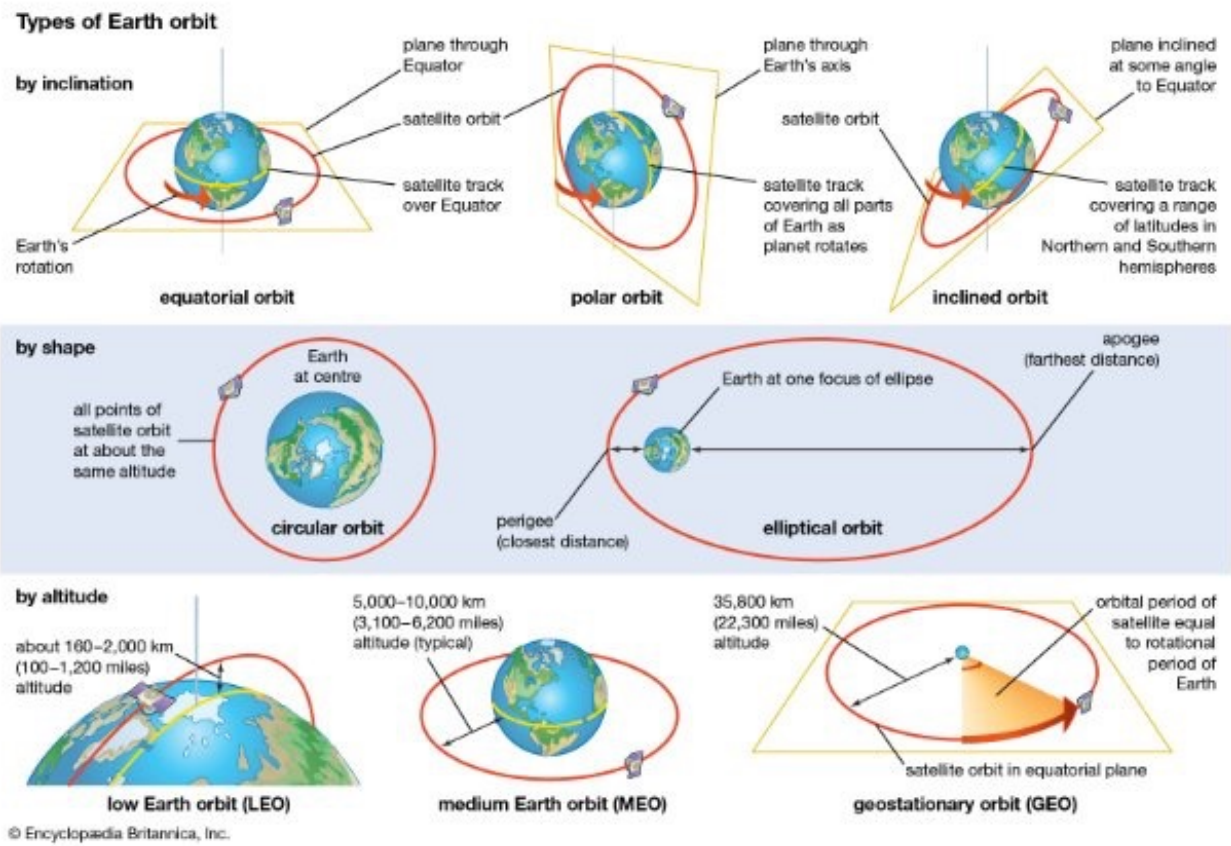


Orbit Type	Geostationary Orbit (GEO)
Altitude	35,786 km above Earth's equator
Orbit Characteristics	GEO satellites appear stationary relative to Earth, orbiting at the same rate as Earth's rotation. Ideal for communication and weather observation.
Usage	Communication satellites, weather satellites, broadcasting satellites
Advantages	Continuous coverage of specific areas, ideal for broadcasting and weather
Disadvantages	Longer signal delays, requires significant energy to reach and maintain position



Types of Orbits Based on Inclination/shape/Functionality				
Orbit Type	**Polar Orbit**	**Sun-Synchronous Orbit (SSO)**	**Inclined Orbit**	**Elliptical Orbit**
Orbit Characteristics	A satellite passes over Earth's poles, covering the entire surface over time as Earth rotates beneath it. Typically a low Earth orbit.	A polar orbit where the satellite's angle with respect to the Sun is fixed, ensuring consistent lighting conditions.	The satellite's orbit is tilted relative to Earth's equator, often used for communication satellites at different latitudes.	Orbits with varying altitudes, increasing in velocity near Earth and decreasing as it moves away, offering diverse mission designs.
Usage	Earth observation satellites, scientific satellites, environmental monitoring (weather, climate, agriculture).	Earth observation satellites needing consistent observation at the same local solar time, for environmental studies.	Satellites for communication or remote sensing, covering regions GEO satellites cannot.	Satellites requiring variable coverage or flexible observation for high-resolution images or communications.
Advantages	Provides global coverage, useful for mapping and weather observation.	Ideal for precise and repetitive observation with consistent lighting conditions.	Offers flexibility in terms of the area covered and allows satellites to serve various latitudes.	Offers varying speeds and altitudes, useful for specific purposes like high-resolution imagery or communication flexibility.
Disadvantages	Higher speed, requiring frequent adjustments and energy consumption.	Requires exact alignment with Earth's precession, making the launch complex.	Less stable than geostationary orbits, requiring more frequent adjustments.	Requires managing high-speed and low-speed movements, and satellite conditions may impact communication and observation at certain altitudes.

Satellite Communication




Thank You All